

C3MAT1T1 – O – 25 – 4213



THIRD SEMESTER B.A./B.SC. (REVISED NEP) DEGREE
EXAMINATION, JANUARY 2026

MATHEMATICS

Algebra II and Real Analysis – I (DSC)

Time : 3 Hours]

[Max. Marks : 80

Instruction : Answer **all** questions.

I. Answer **all** questions of the following :

(10×2=20)

- 1) Show that the identity element of a group is unique.
- 2) Define the left and right cosets of subgroup H of group G.
- 3) Define transitive relation and give an example.
- 4) Show that set of all integers is countable.
- 5) Define limit of a sequence.
- 6) Find $\lim_{n \rightarrow \infty} (n!)^{1/n}$.
- 7) State Cauchy's general principle of convergence.
- 8) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$.
- 9) State Cauchy's root test for series of positive terms.
- 10) Test the convergence of $1 - \frac{1}{4} + \frac{1}{9} - \frac{1}{16} + \dots$

II. Answer **any six** of the following :

(6×5=30)

- 11) State and prove Lagrange's theorem for a finite group.
- 12) Prove that a non empty subset H of a group G is a subgroup of G if and only if $ab^{-1} \in H$ whenever $a, b \in H$.
- 13) Prove that unit interval $[0, 1]$ is uncountable.
- 14) With usual notation, show that $N \times N$ is countable.

[P.T.O.]



- 13) If $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then prove that $\lim_{x \rightarrow a} [f(x) + g(x)] = l + m$.
- 14) Show that the curves $r = a(1 + \sin\theta)$ and $r = b(1 - \sin\theta)$ cut orthogonally.
- 15) A function $f(x)$ is continuous in a closed interval $[a, b]$, then prove that $f(x)$ attains its bounds at least once.
- 16) Derive the formula for the radius of curvature of the curve in cartesian form.
- 17) State and prove Cauchy's mean value theorem.
- 18) Find the n^{th} derivative of $e^{ax}\cos(bx + c)$.

III. Answer **any three** full questions :

(3×10=30)

- 19) a) Reduce the matrix $\begin{pmatrix} 1 & 0 & 1 & 1 \\ -1 & -2 & 1 & 3 \\ 3 & 2 & 1 & -1 \\ 1 & 4 & 5 & 9 \end{pmatrix}$ to the normal form and then find its rank.
- b) Verify Cayley Hamilton theorem for the matrix $\begin{pmatrix} 2 & 4 \\ 3 & 1 \end{pmatrix}$.
- 20) a) State and prove intermediate value theorem.
- b) Evaluate $\lim_{x \rightarrow 0} \left(\frac{a^x + b^x}{2} \right)^{\frac{1}{x}}$.
- 21) a) With usual notation prove that $\tan\phi = r \frac{d\theta}{dr}$.
- b) Find the pedal equation of the curve $r = a(1 - \cos\theta)$.
- 22) a) State and prove Leibnitz's theorem to find the n^{th} derivative of product of two functions.
- b) If $y = e^{m \sin^{-1} x}$ then prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - (n^2 + m^2)y_n = 0$.
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